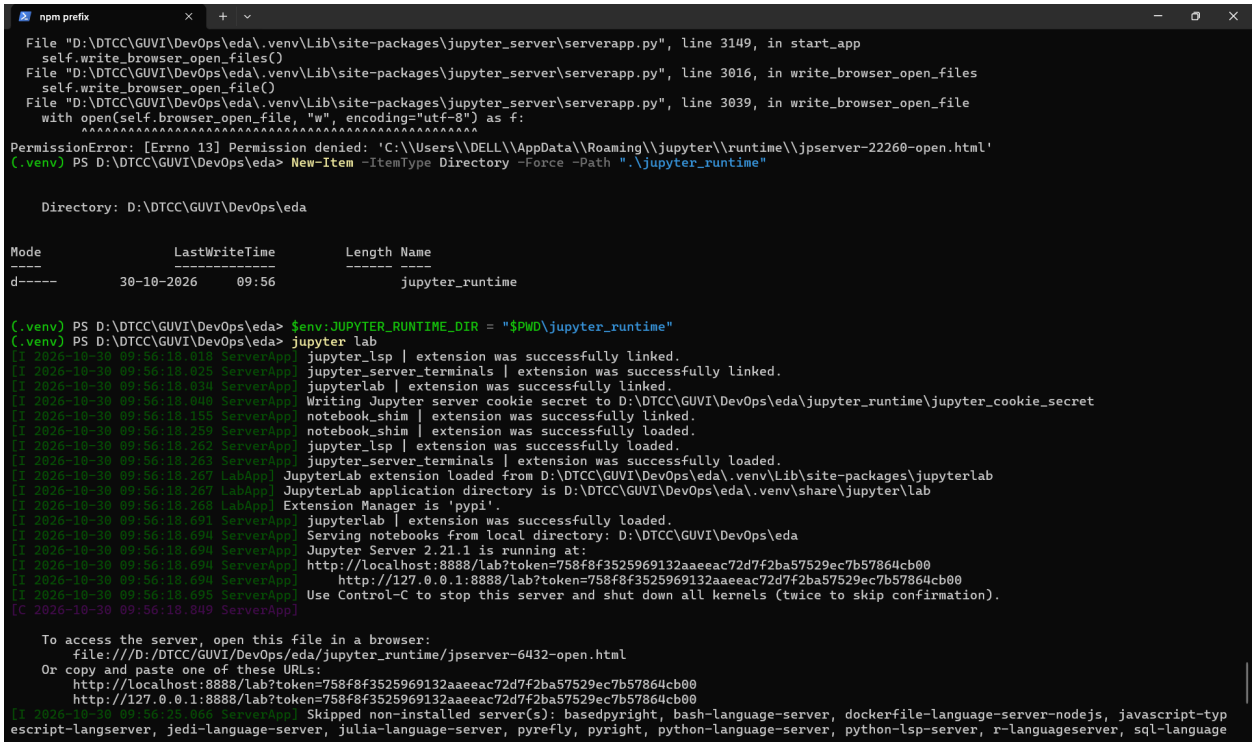
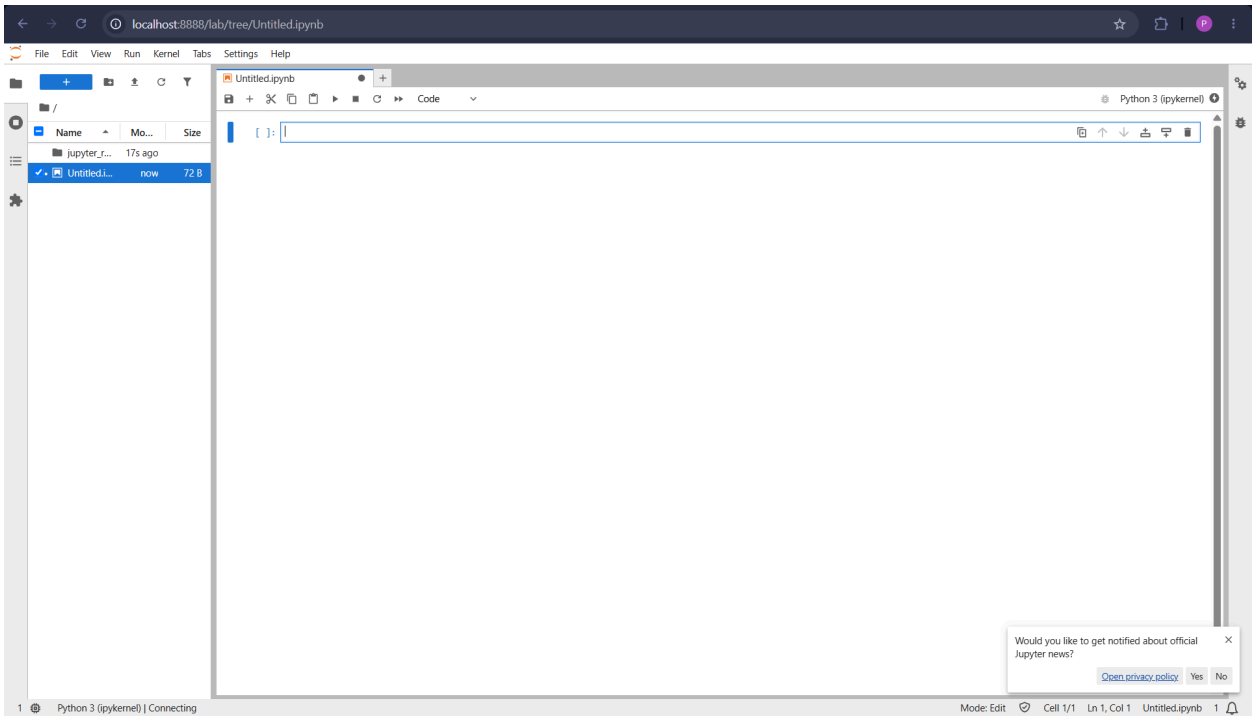
```
npm prefix
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

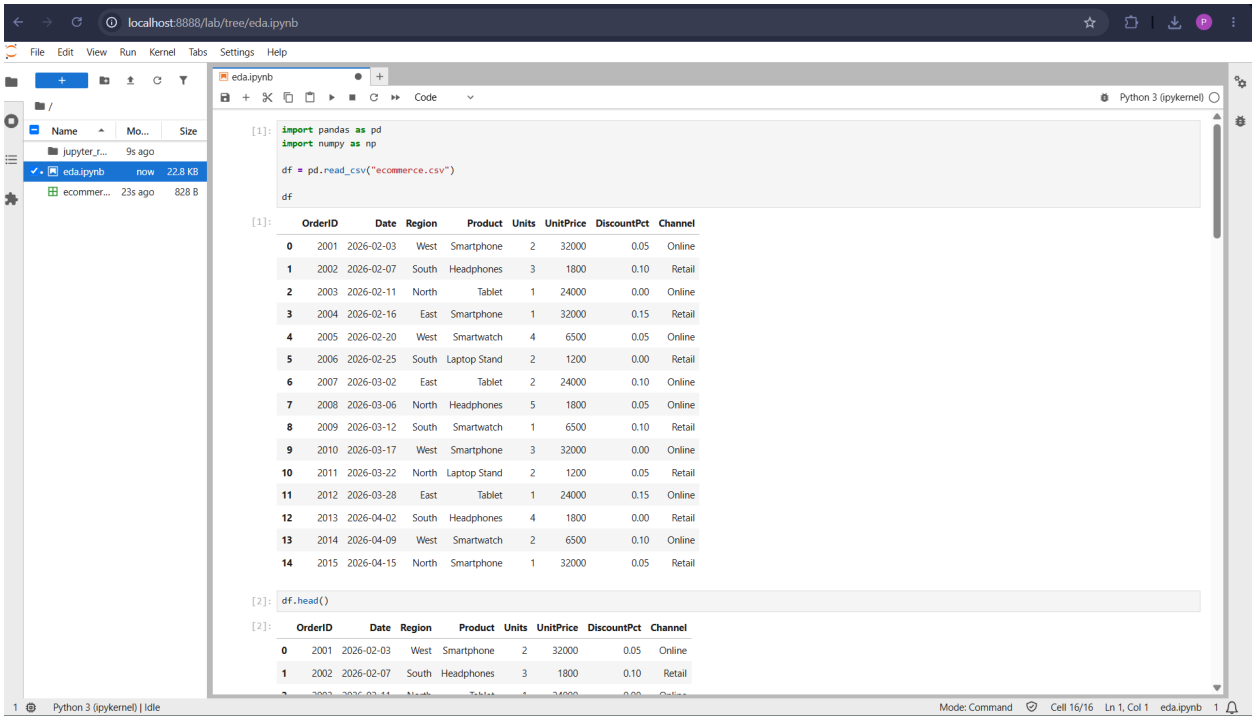
PS D:\DTCC\GUVI\DevOps\eda> .\.venv\Scripts\activate
(.venv) PS D:\DTCC\GUVI\DevOps\eda> python -m pip install pandas numpy matplotlib jupyterlab
Collecting pandas
  Obtaining dependency information for pandas from https://files.pythonhosted.org/packages/10/72/06aefff8fe5efc5e1ca8bfb2695b648a435358f6dc224d4e40159314181a/pandas-3.1.0-cp312-cp312-win_amd64.whl.metadata
  Using cached pandas-3.1.0-cp312-cp312-win_amd64.whl.metadata (18 kB)
Collecting numpy
  Obtaining dependency information for numpy from https://files.pythonhosted.org/packages/12/33/fefe62073dc8acfd0f2b9ed7c003af2f50aa61555e113e6db02b8f79f145/numpy-2.5.4-cp312-cp312-win_amd64.whl.metadata
  Using cached numpy-2.5.4-cp312-cp312-win_amd64.whl.metadata (6.6 kB)
Collecting matplotlib
  Obtaining dependency information for matplotlib from https://files.pythonhosted.org/packages/04/93/0d239bde12b308262265c2d98909b2f0ecad2b5deeae962416422e82e9c7/matplotlib-3.11.2-cp312-cp312-win_amd64.whl.metadata
  Using cached matplotlib-3.11.2-cp312-cp312-win_amd64.whl.metadata (80 kB)
Collecting jupyterlab
  Obtaining dependency information for jupyterlab from https://files.pythonhosted.org/packages/41/8f/58a539ef036aca50891a06b283f044da84a6544177fb6e49217c5c5206c9/jupyterlab-4.6.5-py3-none-any.whl.metadata
  Downloading jupyterlab-4.6.5-py3-none-any.whl.metadata (16 kB)
Collecting python-dateutil>=2.9.0 (from pandas)
  Obtaining dependency information for python-dateutil>=2.9.0 from https://files.pythonhosted.org/packages/5a/17/cebee01d1c0c1441174cf0456ddc91b280d79acb65340c05e41d77866ee8/python_dateutil-3.0.0-py2.py3-none-any.whl.metadata
  Using cached python_dateutil-3.0.0-py2.py3-none-any.whl.metadata (8.2 kB)
Collecting tzdata (from pandas)
  Obtaining dependency information for tzdata from https://files.pythonhosted.org/packages/0a/66/9a06c2227fedfd4919f0eaaa8b94377d9730882a7e4f95823c9a331ab5e1/tzdata-2026.6-py2.py3-none-any.whl.metadata
  Using cached tzdata-2026.6-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting contourpy>=1.0.1 (from matplotlib)
  Obtaining dependency information for contourpy>=1.0.1 from https://files.pythonhosted.org/packages/50/a9/dab08786bb4d77ef9a046b3c1be923be4e73c5d07d91f5a54e8ea9b09e41/contourpy-1.4.0-cp312-cp312-win_amd64.whl.metadata
  Using cached contourpy-1.4.0-cp312-cp312-win_amd64.whl.metadata (3.8 kB)
Collecting cycler>=0.10 (from matplotlib)
  Obtaining dependency information for cycler>=0.10 from https://files.pythonhosted.org/packages/e7/05/c19819d5e3d95294a6f5947fb9b9629efb316b96de511b418c53d245aae6/cycler-0.12.1-py3-none-any.whl.metadata
  Using cached cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.28.2 (from matplotlib)
  Obtaining dependency information for fonttools>=4.28.2 from https://files.pythonhosted.org/packages/1c/3e/ef786310a3774ae7aacaa288ca6d1e1dfd03c242141374784e4b630b76af0/fonttools-4.68.0-cp312-cp312-win_amd64.whl.metadata
  Using cached fonttools-4.68.0-cp312-cp312-win_amd64.whl.metadata (142 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib)
  Obtaining dependency information for kiwisolver>=1.3.1 from https://files.pythonhosted.org/packages/6e/7c/7b210498f9f92e1cd7855f260fa69ef056881087b199ee20c208f0e4189a/kiwisolver-1.5.1-cp312-cp312-win_amd64.whl.metadata
  Using cached kiwisolver-1.5.1-cp312-cp312-win_amd64.whl.metadata (5.2 kB)
Collecting packaging>=20.0 (from matplotlib)
  Obtaining dependency information for packaging>=20.0 from https://files.pythonhosted.org/packages/63/34/ba1c580383c9eada3711951fef0795c80b829a078d72188184
```

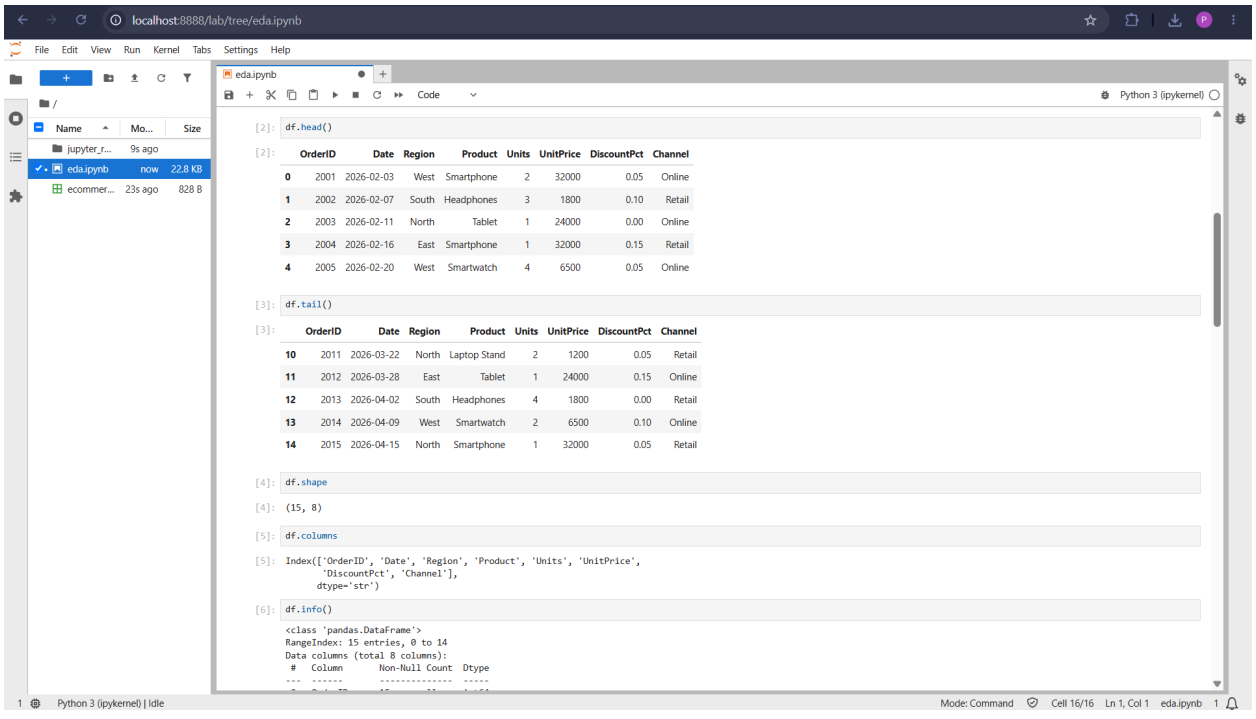
```
npm prefix
ments-2.21.0 pyparsing-3.3.3 python-dateutil-3.0.0 python-json-logger-4.2.0 pywinpty-3.1.2 pyyaml-6.0.3 pyzmq-27.2.0 referencing-0.37.0 requests-2.34.2 rfc3339-validator-0.1.4 rfc3986-validator-0.1.1 rfc3987-syntax-1.1.0 rpds-py-2026.9.1 send2trash-2.1.0 six-1.17.0 soupsieve-3.0.4 stack_data-0.6.3 terminado-0.18.1 tinycss2-1.5.1 tornado-6.6 traitlets-5.16.1 typing_extensions-4.16.0 tzdata-2026.6 uri-template-1.3.0 urllib3-2.8.0 wcwidth-0.9.4 webcolors-25.10.0 webencodings-0.6.1 websocket-client-1.9.2

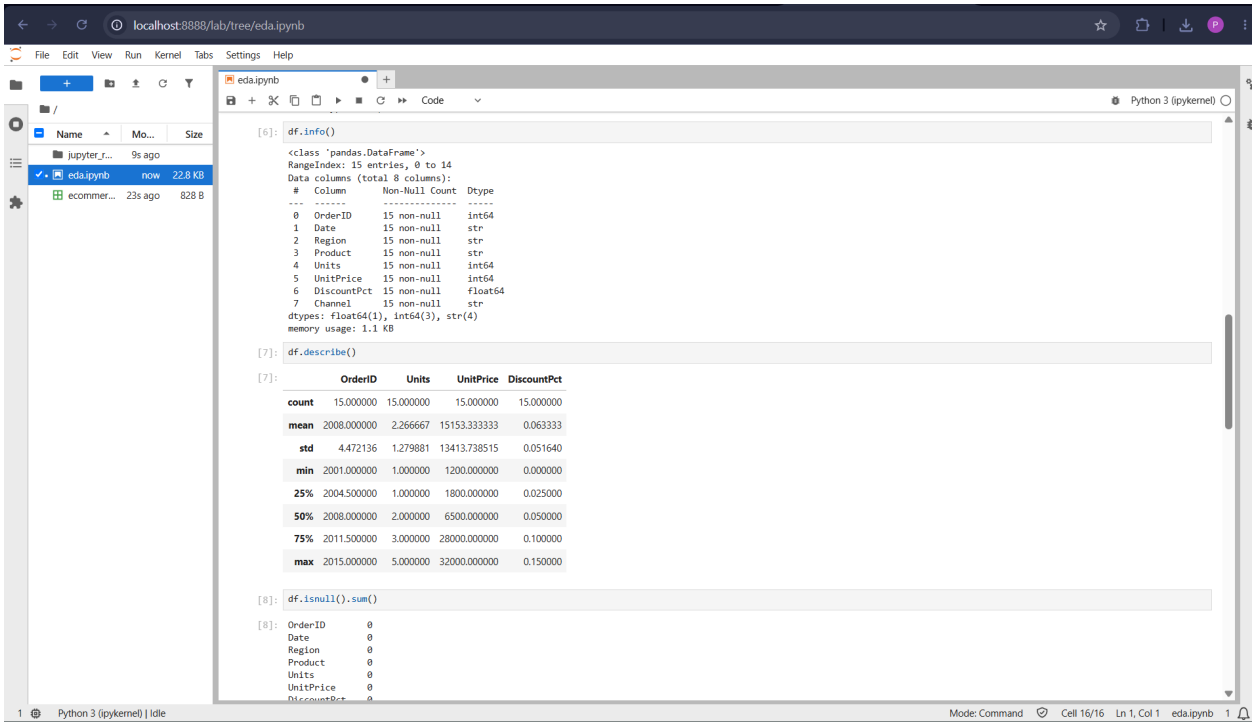
[notice] A new release of pip is available: 23.2.1 -> 26.2.2
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS D:\DTCC\GUVI\DevOps\eda> python -m pip install --upgrade pip
Requirement already satisfied: pip in d:\dtcc\guvi\devops\eda\.venv\lib\site-packages (23.2.1)
Collecting pip
  Obtaining dependency information for pip from https://files.pythonhosted.org/packages/3f/06/3e44203023363fbb71be3c7f35ce042f54bfca330e3e66b3a2e3c4cca579f/pip-26.2.2-py3-none-any.whl.metadata
  Using cached pip-26.2.2-py3-none-any.whl.metadata (4.6 kB)
Using cached pip-26.2.2-py3-none-any.whl (1.8 MB)
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 23.2.1
    Uninstalling pip-23.2.1:
      Successfully uninstalled pip-23.2.1
Successfully installed pip-26.2.2
(.venv) PS D:\DTCC\GUVI\DevOps\eda> jupyter lab
[I 2026-10-30 09:54:42.399 ServerApp] Extension package jupyter_lsp took 0.4720s to import
[I 2026-10-30 09:54:42.871 ServerApp] Extension package jupyter_server_terminals took 0.4708s to import
[I 2026-10-30 09:54:42.873 ServerApp] jupyter_lsp | extension was successfully linked.
[I 2026-10-30 09:54:42.883 ServerApp] jupyter_server_terminals | extension was successfully linked.
[I 2026-10-30 09:54:42.896 ServerApp] jupyterlab | extension was successfully linked.
[I 2026-10-30 09:54:43.131 ServerApp] notebook_shim | extension was successfully linked.
[I 2026-10-30 09:54:43.273 ServerApp] notebook_shim | extension was successfully loaded.
[I 2026-10-30 09:54:43.277 ServerApp] jupyter_lsp | extension was successfully loaded.
[I 2026-10-30 09:54:43.280 ServerApp] jupyter_server_terminals | extension was successfully loaded.
[I 2026-10-30 09:54:43.288 LabApp] JupyterLab extension loaded from D:\DTCC\GUVI\DevOps\eda\.venv\Lib\site-packages\jupyterlab
[I 2026-10-30 09:54:43.288 LabApp] JupyterLab application directory is D:\DTCC\GUVI\DevOps\eda\.venv\share\jupyter\lab
[I 2026-10-30 09:54:43.289 LabApp] Extension Manager is 'pypi'.
[I 2026-10-30 09:54:43.785 ServerApp] jupyterlab | extension was successfully loaded.
[I 2026-10-30 09:54:43.788 ServerApp] Serving notebooks from local directory: D:\DTCC\GUVI\DevOps\eda
[I 2026-10-30 09:54:43.788 ServerApp] Jupyter Server 2.21.1 is running at:
[I 2026-10-30 09:54:43.788 ServerApp] http://localhost:8888/lab?token=48e71b35893061e8d1c4dd09ba83d9f17cf09cd1857bc07b
[I 2026-10-30 09:54:43.788 ServerApp]     http://127.0.0.1:8888/lab?token=48e71b35893061e8d1c4dd09ba83d9f17cf09cd1857bc07b
[I 2026-10-30 09:54:43.788 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
[E 2026-10-30 09:54:43.789 ServerApp] Failed to write server-info to C:\Users\DELL\AppData\Roaming\jupyter\runtime\jpserver-22260.json: PermissionError(13, 'Permission denied')
Traceback (most recent call last):
  File "<frozen runpy>", line 198, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
  File "D:\DTCC\GUVI\DevOps\eda\.venv\Scripts\jupyter-lab.EXE\__main__.py", line 7, in <module>
  File "D:\DTCC\GUVI\DevOps\eda\.venv\Lib\site-packages\jupyter_server\extension\application.py", line 635, in launch_instance
    serverapp.start()
```

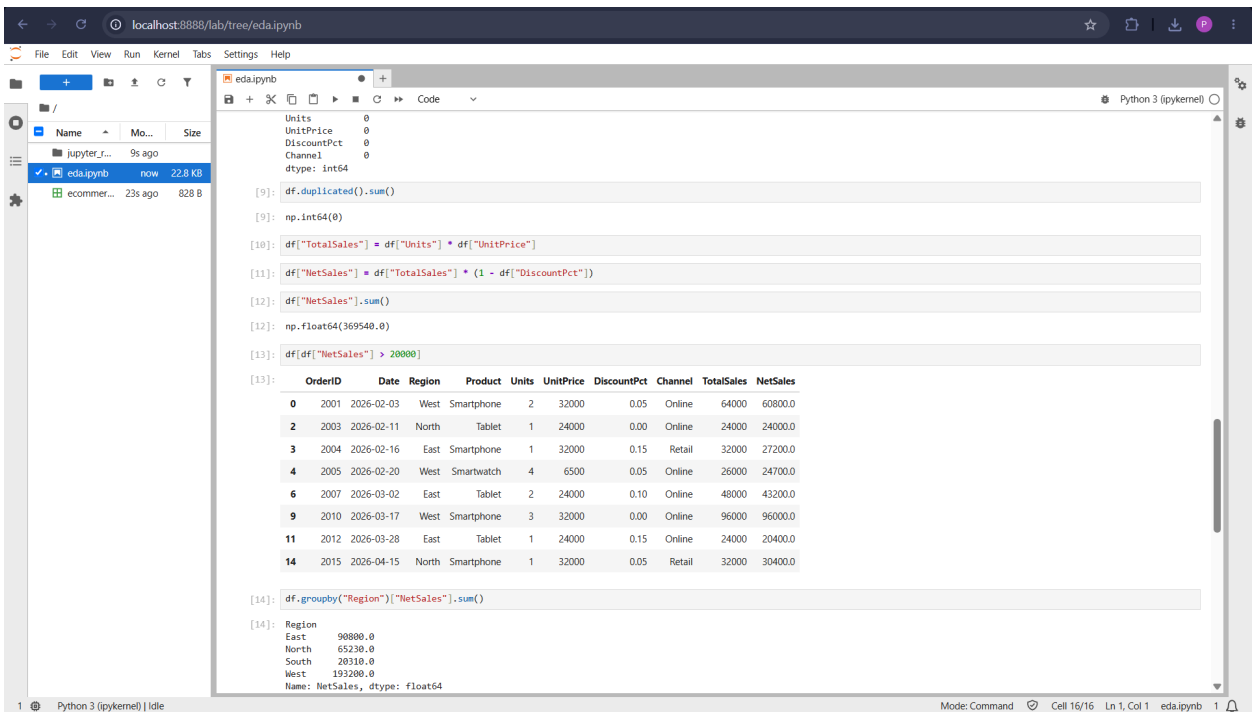

npm prefix
File "D:\DTCC\GUVI\DevOps\eda\.venv\Lib\site-packages\jupyter_server\serverapp.py", line 3149, in start_app
self.write_browser_open_files()
File "D:\DTCC\GUVI\DevOps\eda\.venv\Lib\site-packages\jupyter_server\serverapp.py", line 3016, in write_browser_open_files
self.write_browser_open_file()
File "D:\DTCC\GUVI\DevOps\eda\.venv\Lib\site-packages\jupyter_server\serverapp.py", line 3039, in write_browser_open_file
with open(self.browser_open_file, "w", encoding="utf-8") as f:
PermissionError: [Errno 13] Permission denied: 'C:\\Users\\DELL\\AppData\\Roaming\\jupyter\\runtime\\jpserver-22260-open.html'
(.venv) PS D:\DTCC\GUVI\DevOps\eda> New-Item -ItemType Directory -Force -Path ".\jupyter_runtime"
Directory: D:\DTCC\GUVI\DevOps\eda
Mode LastWriteTime Length Name
d----- 30-10-2026 09:56 jupyter_runtime
(.venv) PS D:\DTCC\GUVI\DevOps\eda> $env:JUPYTER_RUNTIME_DIR = "$PWD\jupyter_runtime"
(.venv) PS D:\DTCC\GUVI\DevOps\eda> jupyter lab
[I 2026-10-30 09:56:18.018 ServerApp] jupyter_lsp | extension was successfully linked.
[I 2026-10-30 09:56:18.025 ServerApp] jupyter_server_terminals | extension was successfully linked.
[I 2026-10-30 09:56:18.034 ServerApp] jupyterlab | extension was successfully linked.
[I 2026-10-30 09:56:18.040 ServerApp] Writing Jupyter server cookie secret to D:\DTCC\GUVI\DevOps\eda\jupyter_runtime\jupyter_cookie_secret
[I 2026-10-30 09:56:18.155 ServerApp] notebook_shim | extension was successfully linked.
[I 2026-10-30 09:56:18.259 ServerApp] notebook_shim | extension was successfully loaded.
[I 2026-10-30 09:56:18.262 ServerApp] jupyter_lsp | extension was successfully loaded.
[I 2026-10-30 09:56:18.263 ServerApp] jupyter_server_terminals | extension was successfully loaded.
[I 2026-10-30 09:56:18.267 LabApp] JupyterLab extension loaded from D:\DTCC\GUVI\DevOps\eda\.venv\Lib\site-packages\jupyterlab
[I 2026-10-30 09:56:18.267 LabApp] JupyterLab application directory is D:\DTCC\GUVI\DevOps\eda\.venv\share\jupyter\lab
[I 2026-10-30 09:56:18.268 LabApp] Extension Manager is 'pypi'.
[I 2026-10-30 09:56:18.691 ServerApp] jupyterlab | extension was successfully loaded.
[I 2026-10-30 09:56:18.694 ServerApp] Serving notebooks from local directory: D:\DTCC\GUVI\DevOps\eda
[I 2026-10-30 09:56:18.694 ServerApp] Jupyter Server 2.21.1 is running at:
[I 2026-10-30 09:56:18.694 ServerApp] http://localhost:8888/lab?token=758f8f3525969132aaeeac72d7f2ba57529ec7b57864cb00
[I 2026-10-30 09:56:18.694 ServerApp] http://127.0.0.1:8888/lab?token=758f8f3525969132aaeeac72d7f2ba57529ec7b57864cb00
[I 2026-10-30 09:56:18.695 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
[C 2026-10-30 09:56:18.849 ServerApp]
To access the server, open this file in a browser:
file:///D:/DTCC/GUVI/DevOps/eda/jupyter_runtime/jpserver-6432-open.html
Or copy and paste one of these URLs:
http://localhost:8888/lab?token=758f8f3525969132aaeeac72d7f2ba57529ec7b57864cb00
http://127.0.0.1:8888/lab?token=758f8f3525969132aaeeac72d7f2ba57529ec7b57864cb00
[I 2026-10-30 09:56:25.066 ServerApp] Skipped non-installed server(s): basedpyright, bash-language-server, dockerfile-language-server-nodejs, javascript-typescript-langserver, jedi-language-server, julia-language-server, pyrefly, pyright, python-language-server, python-lsp-server, r-languageserver, sql-language

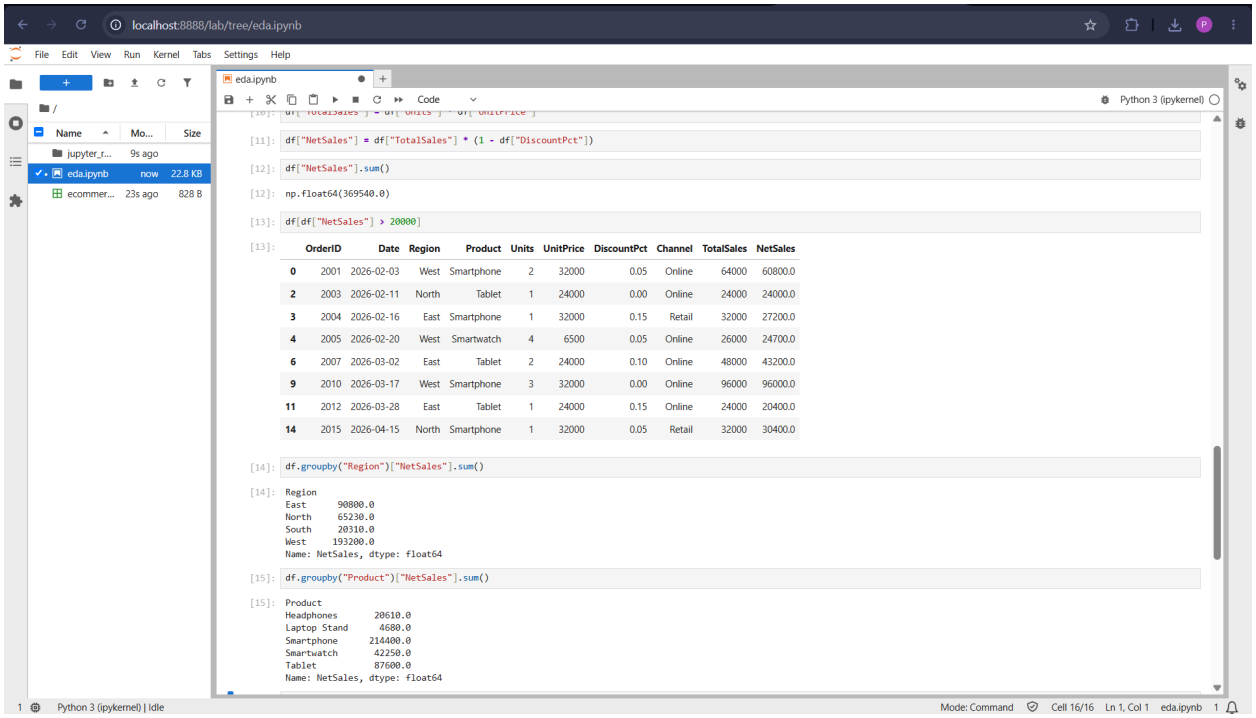

localhost:8888/lab/tree/Untitled.ipynb
File Edit View Run Kernel Tabs Settings Help
Untitled.ipynb
Code
Python 3 (ipykernel)
Name
Mo...
Size
jupyter_r...
17s ago
Untitled.i...
now
72 B
Would you like to get notified about official Jupyter news?
Open privacy policy
Yes
No
Python 3 (ipykernel) | Connecting
Mode: Edit
Cell 1/1
Ln 1, Col 1
Untitled.ipynb


localhost:8888/lab/tree/eda.ipynb
File Edit View Run Kernel Tabs Settings Help
eda.ipynb
Code
Python 3 (ipykernel)
Name Mo... Size
jupyter_r... 9s ago
eda.ipynb now 22.8 KB
ecommer... 23s ago 828 B
[1]:
import pandas as pd
import numpy as np
df = pd.read_csv("ecommerce.csv")
df
[1]:
OrderID Date Region Product Units UnitPrice DiscountPct Channel
0 2001 2026-02-03 West Smartphone 2 32000 0.05 Online
1 2002 2026-02-07 South Headphones 3 1800 0.10 Retail
2 2003 2026-02-11 North Tablet 1 24000 0.00 Online
3 2004 2026-02-16 East Smartphone 1 32000 0.15 Retail
4 2005 2026-02-20 West Smartwatch 4 6500 0.05 Online
5 2006 2026-02-25 South Laptop Stand 2 1200 0.00 Retail
6 2007 2026-03-02 East Tablet 2 24000 0.10 Online
7 2008 2026-03-06 North Headphones 5 1800 0.05 Online
8 2009 2026-03-12 South Smartwatch 1 6500 0.10 Retail
9 2010 2026-03-17 West Smartphone 3 32000 0.00 Online
10 2011 2026-03-22 North Laptop Stand 2 1200 0.05 Retail
11 2012 2026-03-28 East Tablet 1 24000 0.15 Online
12 2013 2026-04-02 South Headphones 4 1800 0.00 Retail
13 2014 2026-04-09 West Smartwatch 2 6500 0.10 Online
14 2015 2026-04-15 North Smartphone 1 32000 0.05 Retail
[2]:
df.head()
[2]:
OrderID Date Region Product Units UnitPrice DiscountPct Channel
0 2001 2026-02-03 West Smartphone 2 32000 0.05 Online
1 2002 2026-02-07 South Headphones 3 1800 0.10 Retail
Python 3 (ipykernel) | Idle
Mode: Command
Cell 16/16
Ln 1, Col 1
eda.ipynb


localhost:8888/lab/tree/eda.ipynb
File Edit View Run Kernel Tabs Settings Help
eda.ipynb
Code
Python 3 (ipykernel)
[2]:
df.head()
[2]:
OrderID Date Region Product Units UnitPrice DiscountPct Channel
0 2001 2026-02-03 West Smartphone 2 32000 0.05 Online
1 2002 2026-02-07 South Headphones 3 1800 0.10 Retail
2 2003 2026-02-11 North Tablet 1 24000 0.00 Online
3 2004 2026-02-16 East Smartphone 1 32000 0.15 Retail
4 2005 2026-02-20 West Smartwatch 4 6500 0.05 Online
[3]:
df.tail()
[3]:
OrderID Date Region Product Units UnitPrice DiscountPct Channel
10 2011 2026-03-22 North Laptop Stand 2 1200 0.05 Retail
11 2012 2026-03-28 East Tablet 1 24000 0.15 Online
12 2013 2026-04-02 South Headphones 4 1800 0.00 Retail
13 2014 2026-04-09 West Smartwatch 2 6500 0.10 Online
14 2015 2026-04-15 North Smartphone 1 32000 0.05 Retail
[4]:
df.shape
[4]:
(15, 8)
[5]:
df.columns
[5]:
Index(['OrderID', 'Date', 'Region', 'Product', 'Units', 'UnitPrice',
'DiscountPct', 'Channel'],
dtype='str')
[6]:
df.info()
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 8 columns):
Column Non-Null Count Dtype
Python 3 (ipykernel) | Idle
Mode: Command
Cell 16/16
Ln 1, Col 1
eda.ipynb


localhost:8888/lab/tree/eda.ipynb
File Edit View Run Kernel Tabs Settings Help
eda.ipynb
Code
Python 3 (ipykernel)
Name
Mo...
Size
jupyter_r... 9s ago
eda.ipynb now 22.8 KB
ecommer... 23s ago 828 B
[6]: df.info()
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 8 columns):
Column Non-Null Count Dtype
0 OrderID 15 non-null int64
1 Date 15 non-null str
2 Region 15 non-null str
3 Product 15 non-null str
4 Units 15 non-null int64
5 UnitPrice 15 non-null int64
6 DiscountPct 15 non-null float64
7 Channel 15 non-null str
dtypes: float64(1), int64(3), str(4)
memory usage: 1.1 KB
[7]: df.describe()
[7]:
OrderID Units UnitPrice DiscountPct
count 15.000000 15.000000 15.000000 15.000000
mean 2008.000000 2.266667 15153.333333 0.063333
std 4.472136 1.279881 13413.738515 0.051640
min 2001.000000 1.000000 1200.000000 0.000000
25% 2004.500000 1.000000 1800.000000 0.025000
50% 2008.000000 2.000000 6500.000000 0.050000
75% 2011.500000 3.000000 28000.000000 0.100000
max 2015.000000 5.000000 32000.000000 0.150000
[8]: df.isnull().sum()
[8]: OrderID 0
Date 0
Region 0
Product 0
Units 0
UnitPrice 0
Python 3 (ipykernel) | Idle
Mode: Command
Cell 16/16
Ln 1, Col 1
eda.ipynb


localhost:8888/lab/tree/eda.ipynb
File Edit View Run Kernel Tabs Settings Help
eda.ipynb
Code
Python 3 (ipykernel)
Name
Mo...
Size
jupyter_r... 9s ago
eda.ipynb now 22.8 KB
ecommer... 23s ago 828 B
Units 0
UnitPrice 0
DiscountPct 0
Channel 0
dtype: int64
[9]: df.duplicated().sum()
[9]: np.int64(0)
[10]: df["TotalSales"] = df["Units"] * df["UnitPrice"]
[11]: df["NetSales"] = df["TotalSales"] * (1 - df["DiscountPct"])
[12]: df["NetSales"].sum()
[12]: np.float64(369540.0)
[13]: df[df["NetSales"] > 20000]
[13]:
OrderID Date Region Product Units UnitPrice DiscountPct Channel TotalSales NetSales
0 2001 2026-02-03 West Smartphone 2 32000 0.05 Online 64000 60800.0
2 2003 2026-02-11 North Tablet 1 24000 0.00 Online 24000 24000.0
3 2004 2026-02-16 East Smartphone 1 32000 0.15 Retail 32000 27200.0
4 2005 2026-02-20 West Smartwatch 4 6500 0.05 Online 26000 24700.0
6 2007 2026-03-02 East Tablet 2 24000 0.10 Online 48000 43200.0
9 2010 2026-03-17 West Smartphone 3 32000 0.00 Online 96000 96000.0
11 2012 2026-03-28 East Tablet 1 24000 0.15 Online 24000 20400.0
14 2015 2026-04-15 North Smartphone 1 32000 0.05 Retail 32000 30400.0
[14]: df.groupby("Region")["NetSales"].sum()
[14]: Region
East 90800.0
North 65230.0
South 20310.0
West 193200.0
Name: NetSales, dtype: float64
Python 3 (ipykernel) | Idle
Mode: Command
Cell 16/16
Ln 1, Col 1
eda.ipynb



OrderID	Date	Region	Product	Units	UnitPrice	DiscountPct	Channel	TotalSales	NetSales
0	2001	2026-02-03	West	Smartphone	2	32000	0.05	Online	64000	60800.0
2	2003	2026-02-11	North	Tablet	1	24000	0.00	Online	24000	24000.0
3	2004	2026-02-16	East	Smartphone	1	32000	0.15	Retail	32000	27200.0
4	2005	2026-02-20	West	Smartwatch	4	6500	0.05	Online	26000	24700.0
6	2007	2026-03-02	East	Tablet	2	24000	0.10	Online	48000	43200.0
9	2010	2026-03-17	West	Smartphone	3	32000	0.00	Online	96000	96000.0
11	2012	2026-03-28	East	Tablet	1	24000	0.15	Online	24000	20400.0
14	2015	2026-04-15	North	Smartphone	1	32000	0.05	Retail	32000	30400.0

Convert the Date column and create Month

```
df["Date"] = pd.to_datetime(df["Date"])
```

```
df["Month"] = df["Date"].dt.to_period("M").astype(str)
```

```
df[["OrderID", "Date", "Month"]]
```

	OrderID	Date	Month
0	2001	2026-02-03	2026-02
1	2002	2026-02-07	2026-02
2	2003	2026-02-11	2026-02
3	2004	2026-02-16	2026-02
4	2005	2026-02-20	2026-02
5	2006	2026-02-25	2026-02
6	2007	2026-03-02	2026-03
7	2008	2026-03-06	2026-03
8	2009	2026-03-12	2026-03
9	2010	2026-03-17	2026-03
10	2011	2026-03-22	2026-03
11	2012	2026-03-28	2026-03
12	2013	2026-04-02	2026-04
13	2014	2026-04-09	2026-04
14	2015	2026-04-15	2026-04

Calculate Revenue

```
df["Revenue"] = (
    df["Units"]
    * df["UnitPrice"]
    * (1 - df["DiscountPct"])
)
```

```
df[["OrderID", "Units", "UnitPrice", "DiscountPct", "Revenue"]]
```

	OrderID	Units	UnitPrice	DiscountPct	Revenue
0	2001	2	32000	0.05	60800.0
1	2002	3	1800	0.10	4860.0
2	2003	1	24000	0.00	24000.0
3	2004	1	32000	0.15	27200.0
4	2005	4	6500	0.05	24700.0
5	2006	2	1200	0.00	2400.0
6	2007	2	24000	0.10	43200.0
7	2008	5	1800	0.05	8550.0
8	2009	1	6500	0.10	5850.0
9	2010	3	32000	0.00	96000.0
10	2011	2	1200	0.05	2280.0
11	2012	1	24000	0.15	20400.0
12	2013	4	1800	0.00	7200.0
13	2014	2	6500	0.10	11700.0
14	2015	1	32000	0.05	30400.0

Region-wise revenue summary

```
region_summary = (
    df.groupby("Region")
    .agg(
        TotalRevenue=("Revenue", "sum"),
        Orders=("OrderID", "count"),
        AverageRevenuePerOrder=("Revenue", "mean")
    )
    .sort_values("TotalRevenue", ascending=False)
)

region_summary
```

Region	TotalRevenue	Orders	AverageRevenuePerOrder
West	193200.0	4	48300.000000
East	90800.0	3	30266.666667
North	65230.0	4	16307.500000
South	20310.0	4	5077.500000

Product-wise revenue summary

```
product_summary = (
    df.groupby("Product")
    .agg(
        TotalRevenue=("Revenue", "sum"),
        UnitsSold=("Units", "sum"),
        Orders=("OrderID", "count")
    )
    .sort_values("TotalRevenue", ascending=False)
)

product_summary
```

Product	TotalRevenue	UnitsSold	Orders
Smartphone	214400.0	7	4
Tablet	87600.0	4	3
Smartwatch	42250.0	7	3
Headphones	20610.0	12	3
Laptop Stand	4680.0	4	2

Monthly revenue summary

```
monthly_summary = (
    df.groupby("Month")
    .agg(
        TotalRevenue=("Revenue", "sum"),
        Orders=("OrderID", "count")
    )
    .sort_index()
)

monthly_summary
```

Month	TotalRevenue	Orders
2026-02	143960.0	6
2026-03	176280.0	6
2026-04	49300.0	3

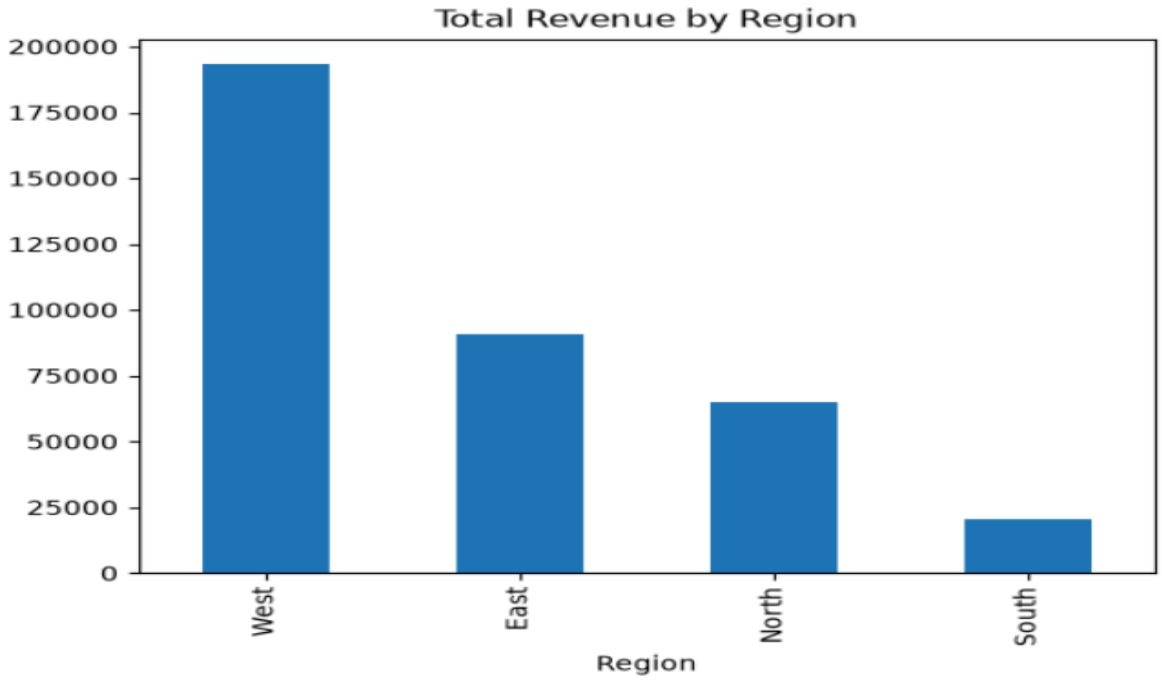
Plot total revenue by region

```
region_summary["TotalRevenue"].plot(
    kind="bar",
    title="Total Revenue by Region"
)

plt.xlabel("Region")
plt.ylabel("Revenue (₹)")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[24], line 6
      2     kind="bar",
      3     title="Total Revenue by Region"
      4 )
      5
----> 6 plt.xlabel("Region")
      7 plt.ylabel("Revenue (₹)")
      8 plt.xticks(rotation=0)
      9 plt.tight_layout()

NameError: name 'plt' is not defined
```



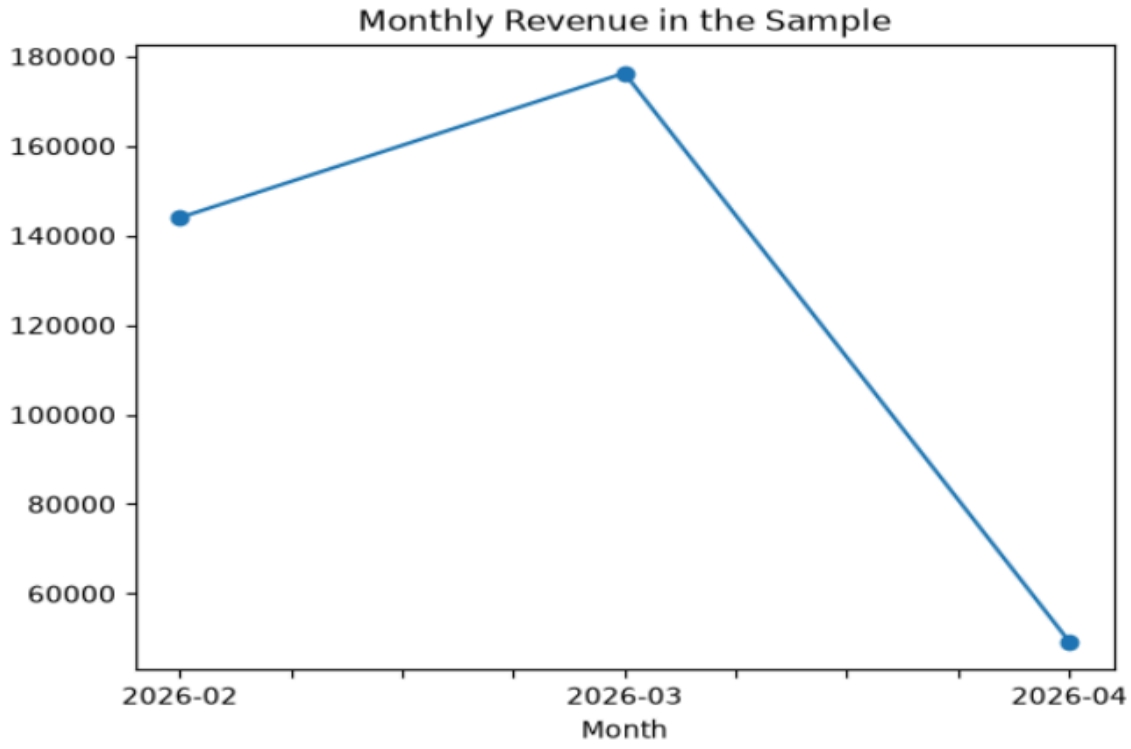
Plot monthly revenue

```
monthly_summary["TotalRevenue"].plot(
    kind="line",
    marker="o",
    title="Monthly Revenue in the Sample"
)

plt.xlabel("Month")
plt.ylabel("Revenue (₹)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[25], line 7
      3     marker="o",
      4     title="Monthly Revenue in the Sample"
      5 )
      6
----> 7 plt.xlabel("Month")
      8 plt.ylabel("Revenue (₹)")
      9 plt.xticks(rotation=45)
     10 plt.tight_layout()

NameError: name 'plt' is not defined
```



Plot total revenue by product

```
product_summary["TotalRevenue"].plot(
    kind="bar",
    title="Total Revenue by Product"
)

plt.xlabel("Product")
plt.ylabel("Revenue (₹)")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[26], line 6
      2     kind="bar",
      3     title="Total Revenue by Product"
      4 )
      5
----> 6 plt.xlabel("Product")
      7 plt.ylabel("Revenue (₹)")
      8 plt.xticks(rotation=0)
      9 plt.tight_layout()

NameError: name 'plt' is not defined
```

